Amazon Sales Deep-Dive Analytics & Insights

Technical Deep-Dive | Prepared for Management

1. PROFIT MARGIN & COST STRUCTURE

Our cost-margin modeling assumes varying levels of supply chain efficiency and product procurement costs. By category, the margins are distributed as: Books (40%), Toys & Games (30%), Clothing (25%), Sports (25%), Home & Kitchen (20%), and Electronics (15%). Based on this, COGS constitutes 60-85% of list price. Discounts applied at checkout directly subtract from the base margin, making the pricing strategy critical.

2. CUSTOMER SEGMENT PROFILE

Customer Segment	Count of Customers
Active Spender	15,218
High-Value Loyalist	14,483
Occasional Buyer	8,493
One-time Buyer	5,039

Segment Definitions:
- High-Value Loyalist: Customers with 3+ transactions and >= $1,500 total spend.
- Active Spenders: Customers with 2+ transactions.
- Occasional Buyer: Customers with 1 transaction and >= $500 total spend.
- One-time Buyer: Customers with 1 transaction and < $500 total spend.

3. PARETO 80-20 CONCENTRATION ANALYSIS

Out of the 50 unique products cataloged in our dataset, the top 40 products account for approximately 80% of the total revenue. This indicates a high concentration of revenue on a few key drivers, such as high-cost electronics and appliances. Focus should remain on supply chain security and inventory health for these top-performing items.

4. PRICING AND DISCOUNT CAPPING ANALYSIS

A critical warning arises from the Electronics category. Since the base margin is 15% (COGS is 85%), any transaction with a discount of 20%, 25%, or 30% yields a negative net margin. Currently, 22% of all Electronics orders are placed with discounts >= 20%, resulting in a net loss. Capping discounts at a maximum of 10% on Electronics will recover over $1.5M in lost profitability annually.